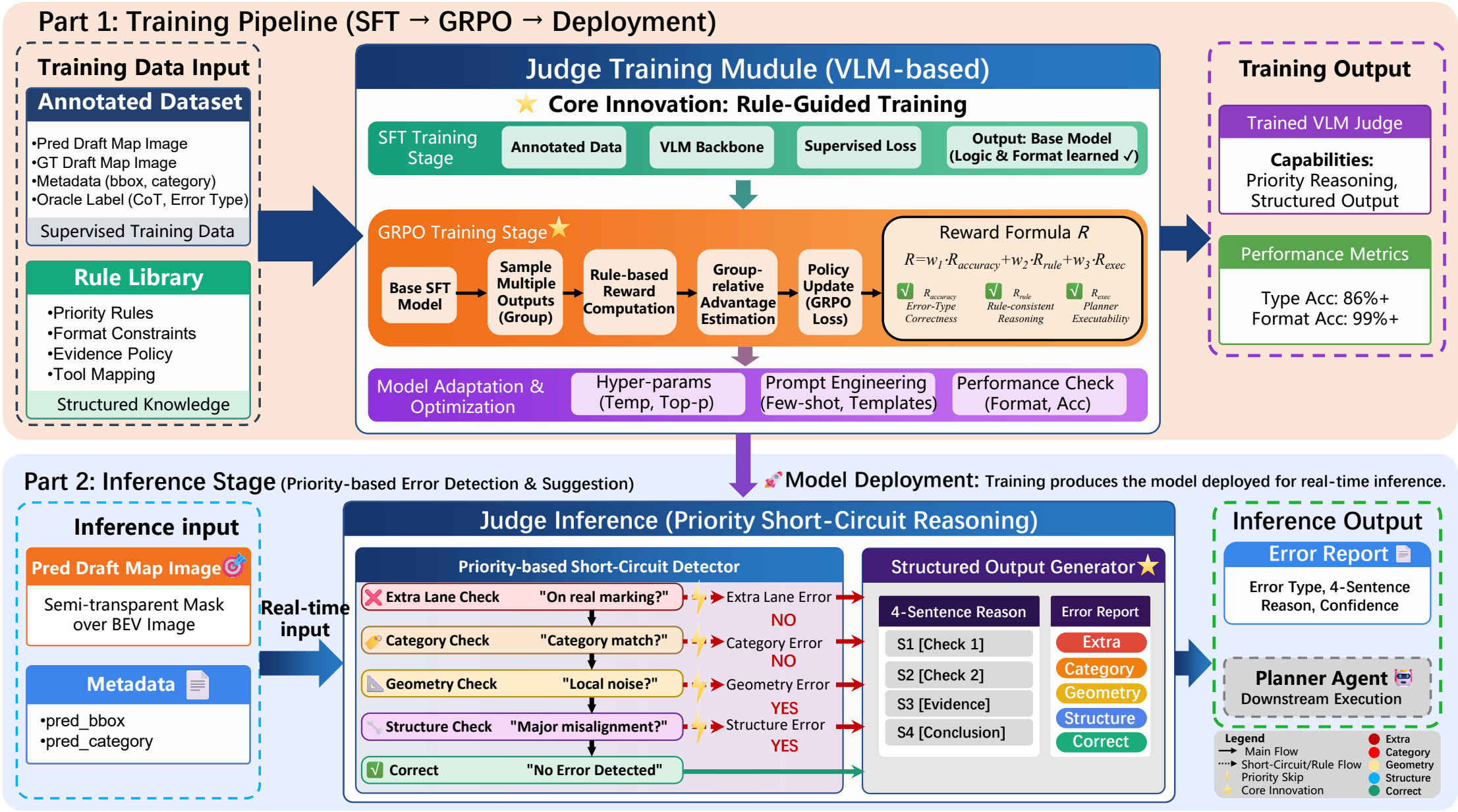
Part 1: Training Pipeline (SFT → GRPO → Deployment)
Training Data Input
Annotated Dataset
•Pred Draft Map Image
•GT Draft Map Image
•Metadata (bbox, category)
•Oracle Label (CoT, Error Type)
Supervised Training Data
Rule Library
•Priority Rules
•Format Constraints
•Evidence Policy
•Tool Mapping
Structured Knowledge
Judge Training Mudule (VLM-based)
Core Innovation: Rule-Guided Training
SFT Training Stage
Annotated Data
VLM Backbone
Supervised Loss
Output: Base Model (Logic & Format learned ✓)
GRPO Training Stage
Base SFT Model
Sample Multiple Outputs (Group)
Rule-based Reward Computation
Group-relative Advantage Estimation
Policy Update (GRPO Loss)
Reward Formula R
$R=w_1 \cdot R_{accuracy}+w_2 \cdot R_{rule}+w_3 \cdot R_{exec}$
$R_{accuracy}$ Error-Type Correctness
R_{rule} Rule-consistent Reasoning
R_{exec} Planner Executability
Model Adaptation & Optimization
Hyper-params (Temp, Top-p)
Prompt Engineering (Few-shot, Templates)
Performance Check (Format, Acc)
Training Output
Trained VLM Judge
Capabilities: Priority Reasoning, Structured Output
Performance Metrics
Type Acc: 86%+
Format Acc: 99%+
Part 2: Inference Stage (Priority-based Error Detection & Suggestion)
Model Deployment: Training produces the model deployed for real-time inference.
Inference input
Pred Draft Map Image
Semi-transparent Mask over BEV Image
Metadata
•pred_bbox
•pred_category
Real-time input
Judge Inference (Priority Short-Circuit Reasoning)
Priority-based Short-Circuit Detector
Extra Lane Check "On real marking?"
Extra Lane Error
NO
Category Check "Category match?"
Category Error
NO
Geometry Check "Local noise?"
Geometry Error
YES
Structure Check "Major misalignment?"
Structure Error
YES
Correct "No Error Detected"
Structured Output Generator
4-Sentence Reason
S1 [Check 1]
S2 [Check 2]
S3 [Evidence]
S4 [Conclusion]
Error Report
Extra
Category
Geometry
Structure
Correct
Inference Output
Error Report
Error Type, 4-Sentence Reason, Confidence
Planner Agent
Downstream Execution
Legend
Main Flow
Short-Circuit/Rule Flow
Priority Skip
Core Innovation
Extra
Category
Geometry
Structure
Correct